

```
% fpga_monitor.m
% MATLAB R2013a — FPGA UART acquisition + filtering + ✓
threshold alert + plot

%% CONFIG
sampleCount      = 20;      % number of samples to collect
fs                = 100;     % sampling rate in Hz
threshold         = 2.0;     % voltage threshold
movWin            = 50;     % moving-average window length
hpCutoff          = 10;     % high-pass cutoff in Hz
notchFreq         = 50;     % notch frequency in Hz
Q                 = 35;     % notch quality factor

%% SERIAL SETUP
portName = 'COM3';          % replace with your COM port
baudRate = 9600;

s = serial(portName, 'BaudRate', baudRate, 'Terminator','LF', ✓
'Timeout',1);
fopen(s);
disp(['Connected to ', portName, ' @ ', num2str(baudRate), ' ✓
baud']);

%% DATA ACQUISITION
rawData = zeros(sampleCount,1);
i = 1;
disp('Collecting samples...');
while i <= sampleCount
    line = fgetl(s);
    if ~ischar(line), continue; end
    val = str2double(line);
    if isnan(val)
        fprintf('[WARNING] skipping: %s\n', line);
        continue;
    end
end
```

```
end
rawData(i) = val;
if val > threshold
    fprintf('Raw: %.3f V -> ☐☐ DANGER!\n', val);
else
    fprintf('Raw: %.3f V -> ☐ ENGINE OK\n', val);
end
i = i + 1;
end

fclose(s);
delete(s);
disp('Serial port closed.');
```

%% FILTER DESIGN

```
% Notch filter
wo = notchFreq/(fs/2);
bw = wo/Q;
[b_notch, a_notch] = iirnotch(wo, bw);
```

% High-pass filter (2nd order Butterworth)

```
[b_hp, a_hp] = butter(2, hpCutoff/(fs/2), 'high');
```

%% SIGNAL PROCESSING

```
data_notched = filtfilt(b_notch, a_notch, rawData);
data_hp = filtfilt(b_hp, a_hp, data_notched);
data_smoothed = filter(ones(movWin,1)/movWin, 1, data_hp);
```

t = (0:sampleCount-1)'/fs;

```
thLine = threshold * ones(size(t));
```

%% PLOTTING

```
figure('Color','w');
plot(t, rawData, 'b-', 'DisplayName','Raw Signal'); hold✓
```

```
on
plot(t, data_smoothed, 'g-', 'LineWidth', 1.5, 'DisplayName', 'Filtered+Smoothed');
plot(t, thLine, 'r--', 'LineWidth', 1.2, 'DisplayName', ['Threshold = ', num2str(threshold), 'V']);
hold off
xlabel('Time (s)');
ylabel('Voltage (V)');
title('FPGA Signal Monitoring & Filtering');
legend('Location', 'best');
grid on;
```